

Computer Architecture- Quiz #4-C

Name: _____ Roll#: _____

Section: _____

Total Time: 20 minutes

Total Marks: 20

7th April, 2026

CLO 2:

Question 1

Marks 10

Consider the following **RISC-V** code. For each instruction, translate it into machine language and represent the result in **binary, hexadecimal** formats. You are also required to clearly identify and label all fields of machine instruction such as **opcode, rd, rs1, rs2, funct3, funct7, and immediate** required for machine tra

Address	Instructions																					
0x00000004	L1: sub t0,t1,t2																					
	<table><tr><td></td><td>Funct7</td><td>RS2</td><td>RS1</td><td>Funct3</td><td>Rd1</td><td>opcode</td></tr><tr><td>Binary</td><td colspan="6"></td></tr><tr><td>Hex</td><td colspan="6"></td></tr></table>		Funct7	RS2	RS1	Funct3	Rd1	opcode	Binary							Hex						
		Funct7	RS2	RS1	Funct3	Rd1	opcode															
	Binary																					
Hex																						
0x00000008	addi s2,s1,12																					
	<table><tr><td></td><td colspan="6">opcode</td></tr><tr><td>Binary</td><td colspan="6"></td></tr><tr><td>Hex</td><td colspan="6"></td></tr></table>		opcode						Binary							Hex						
		opcode																				
	Binary																					
Hex																						
0x0000000C	lw t2,8(s2)																					
	<table><tr><td></td><td colspan="6">opcode</td></tr><tr><td>Binary</td><td colspan="6"></td></tr><tr><td>Hex</td><td colspan="6"></td></tr></table>		opcode						Binary							Hex						
		opcode																				
	Binary																					
Hex																						
0x00000010	sw t2,-6(s2)																					
	<table><tr><td></td><td colspan="6">opcode</td></tr><tr><td>Binary</td><td colspan="6"></td></tr><tr><td>Hex</td><td colspan="6"></td></tr></table>		opcode						Binary							Hex						
		opcode																				
	Binary																					
Hex																						
0x00000014	beq s0,t5,L1																					
	<table><tr><td></td><td colspan="6">opcode</td></tr><tr><td>Binary</td><td colspan="6"></td></tr><tr><td>Hex</td><td colspan="6"></td></tr></table>		opcode						Binary							Hex						
		opcode																				
	Binary																					
Hex																						

CLO 2:**Question 2****Marks 10**

Consider the following machine code and convert it to equivalent **RISC-V** code

Address	Instruction									
0x00000004	0x0124E433 <table border="1"><tr><td></td><td></td><td>opcode</td></tr><tr><td>Binary</td><td colspan="2"></td></tr><tr><td>Decimal</td><td colspan="2"></td></tr></table> Final Instruction:			opcode	Binary			Decimal		
		opcode								
Binary										
Decimal										
0x00000008	0xFF430293 <table border="1"><tr><td></td><td></td><td>opcode</td></tr><tr><td>Binary</td><td colspan="2"></td></tr><tr><td>Decimal</td><td colspan="2"></td></tr></table> Final Instruction:			opcode	Binary			Decimal		
		opcode								
Binary										
Decimal										
0x0000000C	0xFF33AD23 <table border="1"><tr><td></td><td></td><td>opcode</td></tr><tr><td>Binary</td><td colspan="2"></td></tr><tr><td>Decimal</td><td colspan="2"></td></tr></table> Final Instruction:			opcode	Binary			Decimal		
		opcode								
Binary										
Decimal										
0x00000010	0x01042E03 <table border="1"><tr><td></td><td></td><td>opcode</td></tr><tr><td>Binary</td><td colspan="2"></td></tr><tr><td>Decimal</td><td colspan="2"></td></tr></table> Final Instruction:			opcode	Binary			Decimal		
		opcode								
Binary										
Decimal										
0x00000014	0xFE8F18E3 <table border="1"><tr><td></td><td></td><td>opcode</td></tr><tr><td>Binary</td><td colspan="2"></td></tr><tr><td>Decimal</td><td colspan="2"></td></tr></table> Final Instruction:			opcode	Binary			Decimal		
		opcode								
Binary										
Decimal										